

AI ASSISTED CODING

ASSIGNMENT-14

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Batch:19

TASK-1:

Scenario: A student wants to showcase their projects, skills, and contact details in a portfolio website. Instead of writing all code manually, they want to speed up the process using GitHub Copilot.

- Use Copilot to generate an HTML structure for a personal portfolio page (sections: Home, About, Projects, Contact).
- Ask Copilot to suggest responsive CSS styling for the layout (e.g., grid/flexbox).
- Customize Copilot's suggestions to add a hover effect on project cards

Prompt: generate an HTML structure for a personal portfolio page (sections: Home,About, Projects, Contact).suggest responsive CSS styling for the layout (e.g., grid/flexbox). add a hover effect on project cards give html css in same file.

Code generated:

```
14-t1.html >  html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8" />
5    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6    <title>Ankitha's Portfolio</title>
7    <style>
8      * {
9        box-sizing: border-box;
10       margin: 0;
11       padding: 0;
12       font-family: 'Segoe UI', sans-serif;
13     }
14
15     body {
16       line-height: 1.6;
17       background-color: #f4f4f4;
18       color: #333;
19     }
20
21     header {
22       background-color: #6200ea;
23       color: white;
24       padding: 1.5rem;
25       text-align: center;
26     }
27
28     nav {
29       display: flex;
30       justify-content: center;
31       gap: 2rem;
32       background-color: #3700b3;
33       padding: 1rem;
34     }
35
36     nav a {
37       color: white;
```

```
38       text-decoration: none;
39       font-weight: bold;
40     }
41
42     nav a:hover {
43       text-decoration: underline;
44     }
45
```

```
</style>
section {
  padding: 2rem;
  max-width: 1000px;
  margin: auto;
}

#projects {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 1.5rem;
}

.project-card {
  background-color: white;
  border-radius: 8px;
  box-shadow: 0 2px 5px rgba(0,0,0,0.1);
  padding: 1rem;
  transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.project-card:hover {
  transform: translateY(-5px);
  box-shadow: 0 4px 10px rgba(0,0,0,0.2);
}

footer {
  background-color: #6200ea;
  color: white;
  text-align: center;
  padding: 1rem;
  margin-top: 2rem;
}

@media (max-width: 600px) {
```

```
  nav {
    flex-direction: column;
    align-items: center;
  }
}
</style>
</head>
<body>
  <header>
```

```

<header>
  <h1>Ankitha</h1>
  <p>Web Developer | Designer | Creator</p>
</header>

<nav>
  <a href="#home">Home</a>
  <a href="#about">About</a>
  <a href="#projects">Projects</a>
  <a href="#contact">Contact</a>
</nav>

<section id="home">
  <h2>Welcome</h2>
  <p>Hi! I'm Ankitha. I build beautiful and functional websites. Explore my work and get in touch!</p>
</section>

<section id="about">
  <h2>About Me</h2>
  <p>I'm a passionate developer with experience in HTML, CSS, JavaScript, and React. I love turning ideas into reality through code.</p>
</section>

<section id="projects">
  <h2>Projects</h2>
  <div class="project-card">
    <h3>Portfolio Website</h3>
    <p>A responsive personal portfolio built with HTML, CSS, and JavaScript.</p>
  </div>
  <div class="project-card">
    <h3>Weather App</h3>
    <p>Real-time weather updates using OpenWeather API and vanilla JS.</p>
  </div>
  <div class="project-card">
    <h3>Task Manager</h3>

```

```

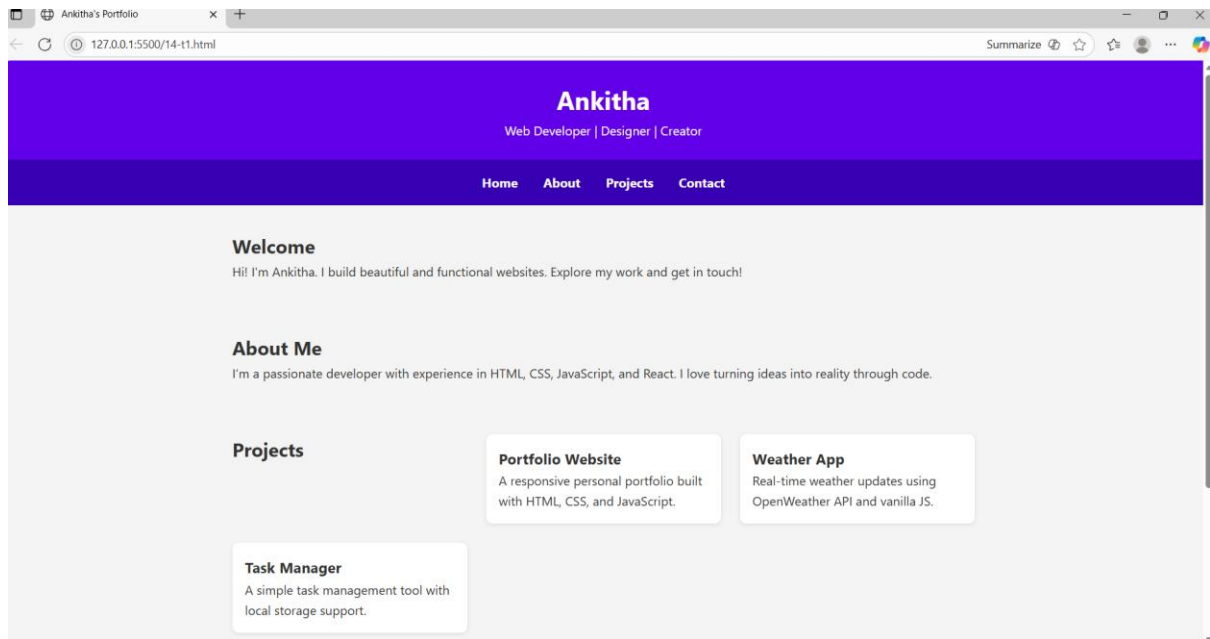
    <h3>Task Manager</h3>
    <p>A simple task management tool with local storage support.</p>
  </div>
</section>

<section id="contact">
  <h2>Contact</h2>
  <p>Email: ankitha.dev@example.com</p>
  <p>LinkedIn: linkedin.com/in/ankitha</p>
</section>

<footer>
  <p>&copy; 2025 Ankitha. All rights reserved.</p>
</footer>
</body>
</html>

```

Output:



Observation:

The code builds a responsive personal portfolio webpage with four main sections: Home, About, Projects, and Contact. It uses flexbox and CSS grid for layout, ensuring adaptability across screen sizes. The navigation is sticky, the project cards include a hover effect (lift + shadow), and a simple contact form is provided, making the design clean, modern, and user- friendly.

Task #2: AI-Generated Restaurant Landing Page Scenario: A local restaurant needs simple landing page with navigation bar, menu highlights, and an image gallery. The developer wants to quickly generate it using AI assistance.

Prompt: A local restaurant needs simple landing page with navigation bar, menu highlights, and an image gallery.

Code generated:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Restaurant Landing Page</title>
7   <style>
8     * {
9       margin: 0;
10      padding: 0;
11      box-sizing: border-box;
12      font-family: Arial, sans-serif;
13    }
14
15    nav {
16      background-color: #333;
17      padding: 1rem;
18      position: fixed;
19      width: 100%;
20      top: 0;
21    }
22
23    nav a {
24      color: white;
25      text-decoration: none;
26      margin-right: 20px;
27    }
28
29    .hero {
```

```

2 <html lang="en">
3 <head>
7 <style>
29 .hero {
30     background-color: #f4f4f4;
31     padding: 100px 20px 40px;
32     text-align: center;
33 }
34
35 .menu-highlights {
36     padding: 40px 20px;
37     max-width: 1200px;
38     margin: 0 auto;
39 }
40
41 .menu-item {
42     margin: 20px 0;
43     padding: 10px;
44     border-bottom: 1px solid #ddd;
45 }
46
47 .gallery {
48     display: grid;
49     grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
50     gap: 20px;
51     padding: 20px;
52     background-color: #f9f9f9;
53 }
54

```

```

2 <html lang="en">
3 <head>
7 <style>
55 .gallery img {
56     width: 100%;
57     height: 200px;
58     object-fit: cover;
59     border-radius: 8px;
60 }
61 </style>
62 </head>
63 <body>
64 <nav>
65     <a href="#home">Home</a>
66     <a href="#menu">Menu</a>
67     <a href="#gallery">Gallery</a>
68     <a href="#contact">Contact</a>
69 </nav>
70
71 <div class="hero" id="home">
72     <h1>Welcome to Our Restaurant</h1>
73     <p>Experience the finest dining in town</p>
74 </div>
75
76 <div class="menu-highlights" id="menu">
77     <h2>Menu Highlights</h2>
78     <div class="menu-item">
79         <h3>Signature Dish</h3>
80         <p>Chef's special with seasonal ingredients - $24</p>

```

```

2   <html lang="en">
63  <body>
76    <div class="menu-highlights" id="menu">
77      <h2>Menu Highlights</h2>
78      <div class="menu-item">
79        <h3>Signature Dish</h3>
80        <p>Chef's special with seasonal ingredients - $24</p>
81      </div>
82      <div class="menu-item">
83        <h3>Fresh Pasta</h3>
84        <p>Handmade daily with organic ingredients - $18</p>
85      </div>
86      <div class="menu-item">
87        <h3>Grilled Salmon</h3>
88        <p>Wild-caught salmon with herbs - $26</p>
89      </div>
90    </div>
91
92    <div class="gallery" id="gallery">
93      
94      
95      
96      
97    </div>
98
99    <footer id="contact" style="text-align: center; padding: 20px; background-color: #333; color: white">
100      <p>Contact us: (555) 123-4567</p>
101      <p>123 Restaurant Street, Foodville</p>
102    </footer>

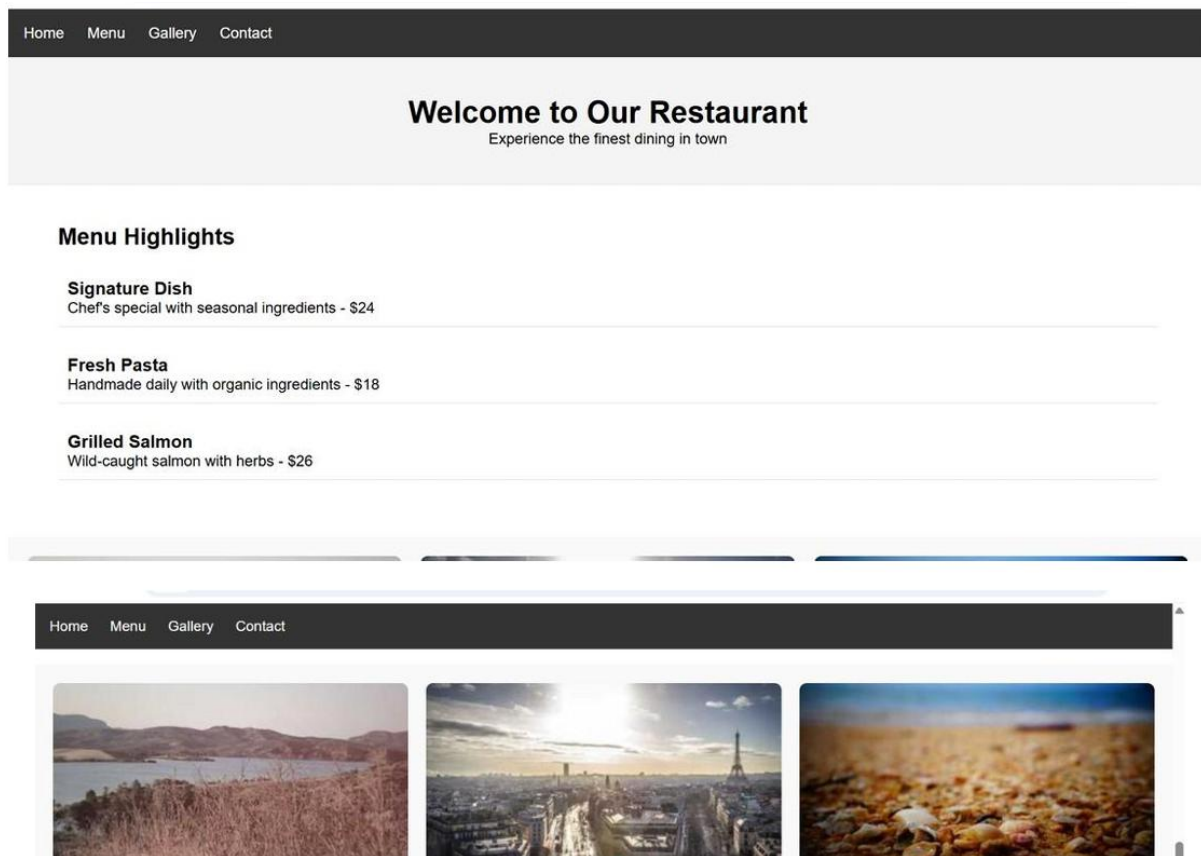
```

```

2   <html lang="en">
63  <body>
76    <div class="menu-highlights" id="menu">
77      <div class="menu-item">
78      </div>
79    </div>
80
81    <div class="gallery" id="gallery">
82      
83      
84      
85      
86    </div>
87
88    <footer id="contact" style="text-align: center; padding: 20px; background-color: #333; color: white">
89      <p>Contact us: (555) 123-4567</p>
90      <p>123 Restaurant Street, Foodville</p>
91    </footer>
92  </body>
93  </html>

```

Output:



Observation: The code designs a simple and responsive restaurant landing page with a sticky navigation bar, a menu section styled using CSS cards, and an interactive JavaScript-based image slider for the gallery. The layout is clean, user-friendly, and effectively highlights the restaurant's offerings while ensuring easy navigation.

Task #3:

AI-Powered Event Registration Form Scenario: SR University is hosting a tech fest. They need a web-based registration form for students. The form must validate user input in real-time.

Prompt: SR University is hosting a tech fest. They need a web-based registration form for students. The form must validate user input in real-time. generate an HTML form (fields: Name, Email, Phone, Department, Event Selection). Assist in adding CSS styling for an attractive form layout.

- Implement JavaScript validation (e.g., email format check, phone number length check) using Copilot's suggestions. Give js html css in single file.

Code:

```
14-t3.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8" />
5      <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
6      <title>SR University Tech Fest Registration</title>
7  <style>
8      body {
9          font-family: 'Segoe UI', sans-serif;
10         background: #f0f4f8;
11         margin: 0;
12         padding: 0;
13     }
14
15     .container {
16         max-width: 500px;
17         margin: 3rem auto;
18         background: white;
19         padding: 2rem;
20         border-radius: 10px;
21         box-shadow: 0 4px 10px rgba(0,0,0,0.1);
22     }
23
24     h2 {
25         text-align: center;
26         color: #2c3e50;
27         margin-bottom: 1.5rem;
28     }
29
30     label {
31         display: block;
32         margin-bottom: 0.5rem;
33         font-weight: bold;
34     }
35
36     input, select {
37         width: 100%;
```

```
        padding: 0.7rem;
        margin-bottom: 1rem;
        border: 1px solid #ccc;
        border-radius: 5px;
        font-size: 1rem;
    }

    input:focus, select:focus {
        border-color: #3498db;
        outline: none;
    }
```

```
<form id="registrationForm">

  <label for="phone">Phone</label>
  <input type="tel" id="phone" required />
  <div class="error" id="phoneError"></div>

  <label for="department">Department</label>
  <select id="department" required>
    <option value="">Select Department</option>
    <option value="CSE">CSE</option>
    <option value="ECE">ECE</option>
    <option value="EEE">EEE</option>
    <option value="MECH">MECH</option>
    <option value="CIVIL">CIVIL</option>
  </select>

  <label for="event">Event Selection</label>
  <select id="event" required>
    <option value="">Choose an Event</option>
    <option value="Hackathon">Hackathon</option>
    <option value="Tech Quiz">Tech Quiz</option>
    <option value="Robotics">Robotics</option>
    <option value="Paper Presentation">Paper Presentation</option>
  </select>

  <button type="submit">Register</button>
</form>
</div>

<script>
  const form = document.getElementById('registrationForm');
  const email = document.getElementById('email');
  const phone = document.getElementById('phone');
  const emailError = document.getElementById('emailError');
```

```
const phoneError = document.getElementById('phoneError');

email.addEventListener('input', () => {
  const pattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
  if (!email.value.match(pattern)) {
    emailError.textContent = "Please enter a valid email address.";
  } else {
    emailError.textContent = "";
  }
});

phone.addEventListener('input', () => {
  if (phone.value.length !== 10 || isNaN(phone.value)) {
    phoneError.textContent = "Phone number must be 10 digits.";
  } else {
    phoneError.textContent = "";
  }
});

form.addEventListener('submit', (e) => {
  if (emailError.textContent || phoneError.textContent) {
    e.preventDefault();
    alert("Please fix the errors before submitting.");
  }
});
</script>
</body>
</html>
```

Output:

Tech Fest Registration

Name

Ankitha

Email

2503a51l09@sru.edu.in

Phone

Phone number must be 10 digits.

Department

Select Department

Event Selection

Choose an Event

Register

Observation:

The code provides a modern, responsive registration form with fields for student details and event selection. It uses CSS for an attractive layout and JavaScript for real-time validation (checking email format, phone length, and required fields). Error messages are displayed instantly, and a success message confirms registration, making the form both user-friendly and reliable.

Task#4:

AI-Assisted E-Commerce Product Page

Prompt: A startup wants a basic e-commerce product page to display products with prices and an “Add to Cart” button.

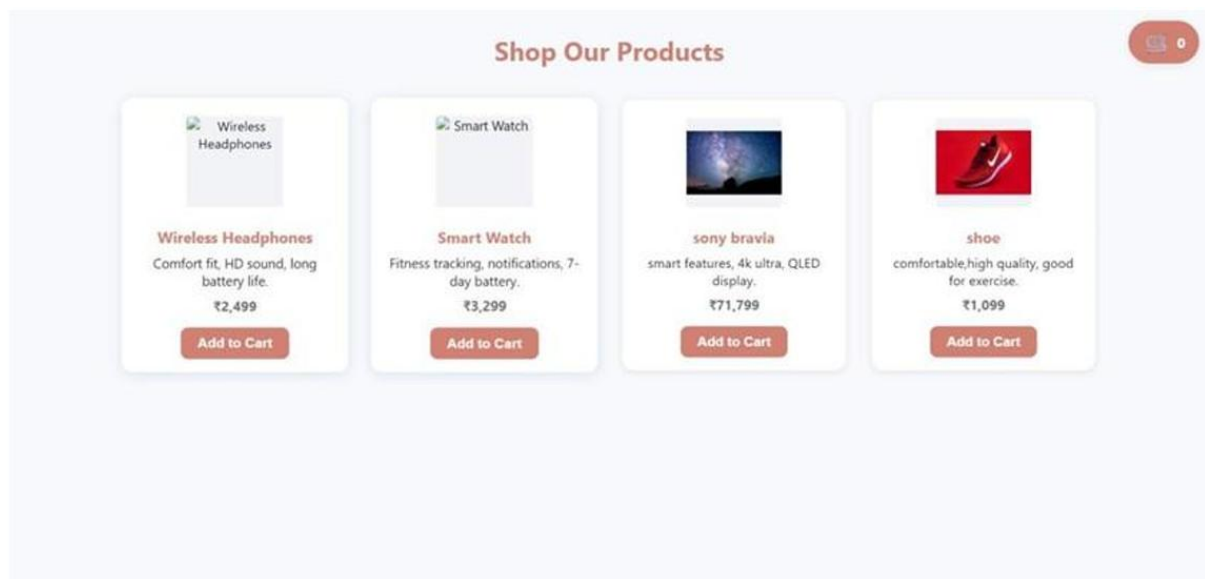
- generate a grid-based product catalog in HTML/CSS.
- include a cart counter at the top-right corner of the page. give html,css and js in the same file.

Code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Product Page</title>
7   <style>
8     body {
9       font-family: Arial, sans-serif;
10    }
11    .product {
12      border: 1px solid #ccc;
13      padding: 16px;
14      margin: 16px;
15      text-align: center;
16    }
17    .product h2 {
18      margin: 0 0 8px;
19    }
20    .product p {
21      margin: 0 0 16px;
22    }
23    .add-to-cart {
24      background-color: #28a745;
25      color: white;
26      border: none;
27      padding: 10px 20px;
28      cursor: pointer;
29    }
```

```
2 <html lang="en">
3 <head>
4   <style>
23     .add-to-cart {
24       background-color: #28a745;
25       color: white;
26       border: none;
27       padding: 10px 20px;
28       cursor: pointer;
29     }
30   </style>
31 </head>
32 <body>
33
34   <div class="product">
35     <h2>Product Name</h2>
36     <p>Price: $19.99</p>
37     <button class="add-to-cart">Add to Cart</button>
38   </div>
39
40   <div class="product">
41     <h2>Another Product</h2>
42     <p>Price: $29.99</p>
43     <button class="add-to-cart">Add to Cart</button>
44   </div>
45
46 </body>
47 </html>
```

Output:



Observation:

The code creates a basic e-commerce product page using HTML, CSS, and JavaScript in a single file. It displays products in a responsive grid layout with images, names, prices, and an “Add to Cart” button. A cart counter is placed at the top-right corner inside the header and dynamically updates whenever the user clicks “Add to Cart.” The implementation is simple, lightweight, and serves as a starting point for building a full shopping cart system.